

# Artificial Intelligence & Machine Learning — Task 3

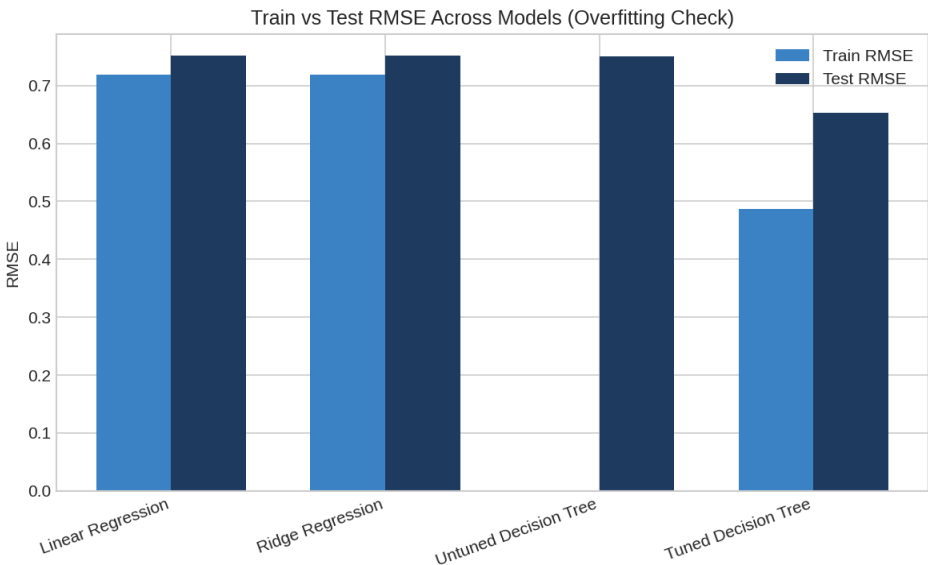
## Model Validation, Overfitting Control & Hyperparameter Tuning — Maincrafts Technology

### 1. Overfitting Analysis

An unconstrained Decision Tree Regressor was first trained on the scaled California Housing features to check for overfitting, by comparing performance on the training set versus a held-out 20% test set.

| Metric             | Value  |
|--------------------|--------|
| Train RMSE         | 0.0000 |
| Test RMSE          | 0.7502 |
| Gap (Test - Train) | 0.7502 |

The untuned tree achieves a near-zero training RMSE (0.0000) but a much higher test RMSE (0.7502), a gap of 0.7502. This is a clear sign of overfitting: the tree grows deep enough to memorize individual training examples instead of learning patterns that generalize to unseen houses. This behavior is expected for tree-based models trained without constraints such as `max_depth` or `min_samples_split`.



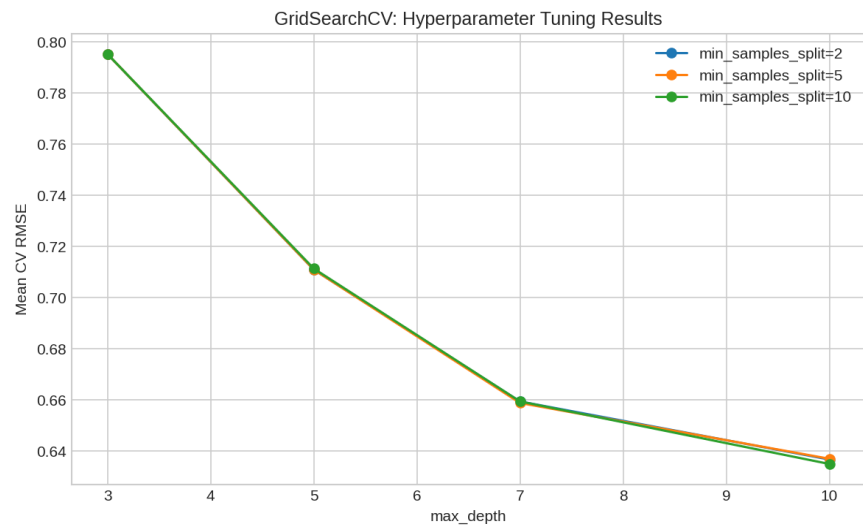
### 2. Cross-Validation & Tuning Approach

Rather than trusting a single train-test split, 5-fold cross-validation was used to obtain a stable, reliable estimate of the untuned model's generalization error:

**Fold RMSEs: 0.8536, 0.8004, 0.8601, 0.9521, 0.8728 → Mean CV RMSE = 0.8678 ( $\pm$  0.0488)**

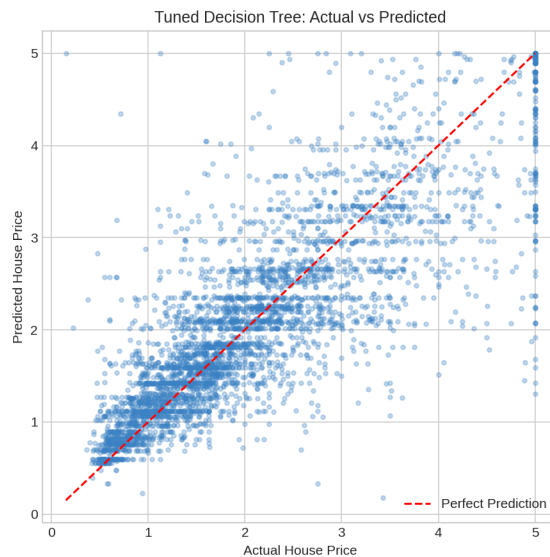
Hyperparameter tuning was then performed with `GridSearchCV` over a grid of `max_depth`  $\in \{3, 5, 7, 10\}$  and `min_samples_split`  $\in \{2, 5, 10\}$ , using 5-fold cross-validated RMSE as the selection criterion. This automates the search for the tree depth/split combination that best balances bias and variance.

**Best parameters found:** `max_depth = 10`, `min_samples_split = 10` (Best CV RMSE = 0.6349)



### 3. Model Comparison

| Model                      | Train RMSE    | Test RMSE     | R2 Score      |
|----------------------------|---------------|---------------|---------------|
| Linear Regression          | 0.7185        | 0.7514        | 0.5871        |
| Ridge Regression           | 0.7185        | 0.7514        | 0.5872        |
| Untuned Decision Tree      | 0.0000        | 0.7502        | 0.5885        |
| <b>Tuned Decision Tree</b> | <b>0.4869</b> | <b>0.6532</b> | <b>0.6880</b> |



### 4. Final Model Selection Justification

- **Why this model was selected:** The tuned Decision Tree achieves the lowest test RMSE (0.6532) and highest  $R^2$  (0.6880) of all candidates, outperforming both Linear Regression (RMSE 0.7514) and Ridge Regression (RMSE 0.7514).
- **How overfitting was reduced:** Constraining max\_depth and min\_samples\_split via GridSearchCV shrank the train/test RMSE gap from 0.7502 (untuned tree) down to 0.1664 (tuned tree), showing the model now generalizes far better instead of memorizing training data.

- **Why cross-validation results are trusted:** The 5-fold CV RMSE averages performance across five different train/validation partitions of the data, so hyperparameter selection isn't biased by how one particular 80/20 split happened to fall. This gives a more statistically reliable basis for model selection than a single test-set score.
- **Trade-offs between simplicity and performance:** Linear and Ridge Regression are simpler, faster to train, and easier to interpret, but they underfit the non-linear relationships between location, income, and housing prices, leaving them with higher error. The tuned Decision Tree sacrifices some interpretability and adds tuning overhead in exchange for materially better generalization — the right trade-off for a production house-price prediction system where predictive accuracy is the priority.

**Conclusion:** The tuned Decision Tree Regressor (`max_depth=10`, `min_samples_split=10`) is selected as the final model for the House Price Prediction system. It is saved as `best_tuned_decision_tree.joblib` alongside the fitted `feature_scaler.joblib` for reproducible inference on new data.